REAL VS. AI-GENERATED AUDIO

Cost-Sensitive Deep Learning Detection and External Generalization

Final Project Report | Group 10 | Deep Learning | August 2026

9,879	177,855	0.972	0.682
SONICS songs	10-second clips	Best SONICS F1	Best external F1

Project objective

Evaluate whether deep-learning models can distinguish real from AI-generated music, and whether strong in-domain performance survives source and generator shift when errors have unequal business costs.

Abstract

This project investigates binary detection of real versus AI-generated music under an explicitly cost-sensitive operating objective. The primary dataset was SONICS: 9,879 source songs (5,000 real and 4,879 AI-generated) standardized to 16 kHz mono audio and segmented into 177,855 non-overlapping 10-second clips. A stratified 70/15/15 split was assigned at the song level before segmentation. During implementation, the original flat clip-pool design was replaced with song-level sampling so that correlated clips from the same recording would not be treated as independent examples or overweight longer songs. Three architectures were compared: a two-layer LSTM trained from scratch on 64-bin log-Mel spectrograms, a frozen PANNs Cnn14_16k backbone with a small classification head, and a frozen MERT-v1-95M music representation model with a small classification head. Manual hyperparameter search was followed by validation-based threshold selection using a normalized business cost in which a false negative was weighted five times more heavily than a false positive. On the SONICS test set, the MERT-based attention model achieved the strongest result (F1=0.972, cost=0.034), followed by the LSTM (F1=0.951, cost=0.069) and Mel-CNN (F1=0.893, cost=0.179). External evaluation on 2,205 GTZAN real-music segments and 2,058 FakeMusicCaps AI-generated clips reversed the ranking: the LSTM generalized best (F1=0.682), while Mel-CNN and MERT dropped to 0.523 and 0.532. The external breakdown showed distinct failure modes, with MERT rarely misclassifying real GTZAN music but missing 60.1% of unfamiliar fake clips. Validation-selected thresholds reduced observed business cost for all three models and outperformed the theoretical 1/6 threshold implied by the cost ratio, demonstrating the practical importance of calibration and empirical operating-point selection. Overall, the results show that strong in-domain accuracy is insufficient evidence of robust AI-music detection: external source coverage, failure-mode analysis, and cost-aware calibration are essential.

Key result in one sentence

The MERT transfer model was best on SONICS, but the scratch LSTM was the strongest external generalizer; the model ranking therefore depended more on domain shift than on in-domain validation strength.

1. Background and Business Motivation

1.1 Industry context

The project was motivated by a rapidly developing business environment around AI-generated music. The accompanying project presentation framed the industry direction around three pressures: streaming platforms increasingly need provenance, disclosure and abuse controls; distributors are being pushed to collect metadata and support automated detection; and regulation and copyright rules are increasing the stakes around transparency, impersonation and rights attribution. The resulting operational question is not simply whether AI music is permitted, but whether platforms can identify undisclosed synthetic content reliably enough to support metadata, moderation and enforcement workflows.

This report treats that landscape as project motivation rather than as a legal analysis. The modeling problem is therefore defined as a binary decision: given a 10-second music excerpt, estimate whether it is real/human-origin music or AI-generated music, while accounting for the fact that the two types of error do not carry equal business consequences.

Pressure area	Project framing
Streaming platforms	Greater emphasis on AI metadata/disclosure, synthetic-spam prevention, fraud controls and impersonation risk.
Distribution and detection	Upstream metadata collection and automated detection can complement artist or distributor declarations.
Regulation and copyright	Transparency, voice-cloning and rights-attribution concerns increase the value of reliable provenance signals.

1.2 Cost-sensitive business problem

The key modeling assumption is that missing AI-generated content is materially more costly than incorrectly flagging human content. A false negative means an AI-generated clip is classified as human; the project associated this with disclosure or compliance failures, copyright or voice-cloning exposure, synthetic-spam leakage, royalty-pool dilution, and reputational or partner risk. A false positive means human music is classified as AI-generated; this can create creator dissatisfaction, manual-review overhead and temporary mislabeling, but was treated as more reversible.

Error	Operational interpretation	Relative cost
False negative	AI-generated music is accepted as human. Potentially allows undisclosed synthetic content, fraud/spam or rights-risk to pass through.	5
False positive	Human music is flagged as AI-generated. Creates review, correction and creator-friction costs.	1

The project therefore used a normalized weighted error cost, where N is the number of evaluated items:

$$\text{Cost} = (\text{FP} + 5\,\text{FN}) \,/\, N$$

Assumed costs: $C_{FP} = 1$, $C_{FN} = 5$

If the model output were a perfectly calibrated probability of the fake class, this cost ratio would imply a Bayes decision threshold of $p^* = 1 / (1 + 5) = 0.167$. In practice, trained neural-network scores are not guaranteed to be calibrated. The project therefore compared this theoretical operating point with a default 0.5 threshold and with a threshold selected empirically on SONICS validation data.

1.3 Research questions

1. How do three architectures with deliberately different inductive biases perform on held-out SONICS data?
2. Does the in-domain ranking remain stable when the models are evaluated on unseen real-music sources and unseen AI generators?
3. How does a 1:5 false-positive/false-negative cost ratio change the preferred decision threshold and the interpretation of model quality?

2. Data and Preprocessing

2.1 SONICS acquisition and manifest

SONICS was the primary training and evaluation source. The real side was downloaded from YouTube using the dataset metadata and converted to 16 kHz mono WAV. The AI-generated side was sampled from the SONICS archives and likewise converted to 16 kHz mono WAV. The final validated manifest contained 9,879 source songs: 5,000 real and 4,879 fake. The fake SONICS material represented the Suno/Udio-generated side used in the project.

The acquisition stage was deliberately separated from the main training notebook because YouTube access required multiple sessions, retries and skip-existing behavior. The main experiment therefore begins from the validated local/Drive audio and manifests rather than repeating acquisition.

2.2 Item-level split before segmentation

A critical correctness decision was to assign the train/validation/test split before any segmentation. Songs were split with a fixed seed (42), stratified by label, using a 70/15/15 ratio. Only after the split was fixed was each recording divided into non-overlapping 10-second clips. Final partial fragments shorter than 10 seconds were dropped. This prevents clips from the same source recording from appearing in multiple partitions.

Split	Songs	Real songs	Fake songs	10-s clips
Train	6,915	3,500	3,415	124,375
Validation	1,482	750	732	26,616
Test	1,482	750	732	26,864
Total	9,879	5,000	4,879	177,855

2.3 Song-level sampling: correcting the unit of analysis

The first dataset implementation treated the 177,855 clips as a flat pool. During execution, this was recognized as a poor statistical unit: a typical song contributes many highly correlated 10-second excerpts that share production style, genre, mastering and generator characteristics. A flat sampler would therefore over-weight songs with more clips and make the effective number of independent examples look much larger than it really is.

The pipeline was changed so that one source song corresponds to one training example. For the compute-constrained HPO stage, a balanced subset of 2,100 training songs was selected (1,050 real and 1,050 fake), with three candidate 10-second clips preselected per song. Each training epoch randomly selected one of those candidates. Validation and test used one fixed middle clip per song, yielding one prediction per source song. The item-level split already protected against cross-partition leakage; the song-level sampler fixed the remaining correlation and weighting problem inside each partition.

> **Why this change matters**
>
> The primary sample size for model comparison is the number of songs, not the number of 10-second fragments. Treating correlated fragments as independent would make results look more certain and would bias training toward recordings that produced more clips.

2.4 External evaluation set

To test out-of-domain generalization, the final pipeline combined GTZAN as real music with FakeMusicCaps as AI-generated music. This differed from the earlier plan, which treated GTZAN as a separate real-only stress test and proposed a MusicCaps/FakeMusicCaps pairing. The combined GTZAN + FakeMusicCaps set was chosen so that the external benchmark contained both classes and could support precision, recall and F1, while the two sources were still analyzed separately to expose error type.

Some GTZAN source files were corrupted and could not be used. The final usable subset contained 735 GTZAN recordings; each approximately 30-second recording was converted to three 10-second segments, producing 2,205 real evaluation items. FakeMusicCaps contributed 2,058 fake 10-second clips. The combined benchmark therefore contained 4,263 items and was near-balanced by class.

External source	Label	Usable source recordings	10-s evaluation items	Role
GTZAN	Real	735	2,205	Out-of-domain real-music / false-positive stress test
FakeMusicCaps	Fake	2,058 clips	2,058	Unseen AI-generator / false-negative stress test
Combined	Both	-	4,263	Secondary binary generalization benchmark

3. Experimental Methodology

3.1 Walking skeleton and scale-up strategy

Before full experimentation, a 100-example walking-skeleton run (50 real and 50 fake) was used to verify the end-to-end path from processed audio through the dataset, model, training loop and evaluation code. The goal was integration correctness rather than performance. After the path was proven, the experiment scaled to the 2,100-song HPO subset and then to the full validation/test partitions.

3.2 Model architectures

The three models were selected to compare distinct representation strategies rather than small variations of one architecture: a scratch sequence model, a general audio-tagging transfer model, and a music-specific pretrained representation model.

Model	Input / backbone	Trainable classifier	Final selected settings
Baseline LSTM	64-bin log-Mel spectrogram; 2-layer unidirectional LSTM trained from scratch	BatchNorm + dropout + linear 2-class head	Hidden=128; dropout=0.5; LR=5e-4; 5 epochs
Mel-CNN	Frozen PANNs Cnn14_16k; 16 kHz waveform -> 2,048-d embedding	Dropout + linear 2-class head	Dropout=0.3; LR=0.01; 10 epochs
Attention / MERT	Frozen m-a-p/MERT-v1-95M; waveform resampled 16 kHz -> 24 kHz; mean pooled hidden states	Dropout + linear 2-class head	Dropout=0.3; LR=0.01; 10 epochs

The PANNs backbone was explicitly changed to the Cnn14_16k checkpoint after a sample-rate mismatch was identified: the default PANNs configuration expected 32 kHz, while the project pipeline standardized audio to 16 kHz. MERT natively expects 24 kHz, so the waveform was resampled from 16 kHz to 24 kHz before inference through the frozen backbone.

3.3 Training protocol and infrastructure

All models used cross-entropy loss and the Adam optimizer with batch size 16. Training ran for a fixed number of epochs while saving both a latest checkpoint and the checkpoint with the best validation accuracy. The transfer models reused precomputed frozen-backbone embeddings, making their hyperparameter sweeps inexpensive compared with end-to-end LSTM training. For repeated reads, selected clips were packed into Drive-resident archives and unpacked to Colab local storage, avoiding severe Drive I/O overhead.

The main HPO subset used 2,100 balanced songs and three candidate clips per song (6,300 training clips available for embedding/cache construction). Validation used all 1,482 validation songs with one fixed crop per song. The seed for sampling and subset construction was 42.

> **Implementation gap: augmentation was not equivalent across models**
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> The Baseline LSTM training path applied dynamic waveform augmentation (Gaussian noise with random SNR and stochastic downsample/upsample bitrate variation) before spectrogram computation. Mel-CNN and MERT were trained from precomputed embeddings generated from unaugmented audio, so their classification heads did not receive the same dynamic augmentation. This is a meaningful comparability limitation and a priority for a rerun.

3.4 Manual hyperparameter search

Manual grid search was used as the primary hyperparameter-selection procedure. The LSTM tested three combinations of hidden dimension, dropout and learning rate; the transfer models tested five learning rates while keeping a 0.3 dropout head. Final checkpoints were selected by validation accuracy within these controlled searches.

Model	Search tried	Selected configuration	Best val. accuracy
Baseline LSTM	3 configs	Hidden 128, dropout 0.5, LR 5e-4	0.924
Mel-CNN	LR: 1e-4, 1e-3, 1e-2, 0.02, 0.05	LR 0.01, dropout 0.3	0.910
Attention / MERT	LR: 1e-4, 1e-3, 1e-2, 0.02, 0.05	LR 0.01, dropout 0.3	0.980

The transfer-model learning-rate sweeps showed a clear low-to-mid-rate optimum followed by a plateau or decline: Mel-CNN peaked at 0.01, while Attention reached the same best validation accuracy at 0.01 and 0.02 and used 0.01 as the final checkpoint. Within the controlled five-epoch LSTM grid, lowering the learning rate to 5e-4 and increasing dropout to 0.5 materially improved validation accuracy over the comparable 1e-3 / 0.3 configuration.

3.5 Supplementary Optuna search

A supplementary 30-trial Optuna search was run for Mel-CNN and Attention, varying learning rate and dropout and minimizing validation business cost rather than validation accuracy. The best recorded validation costs were 0.175 for Mel-CNN (LR approximately 0.0245, dropout approximately 0.103) and 0.025 for Attention (LR approximately 0.0104, dropout approximately 0.254), both lower than the corresponding manually tuned validation costs.

These Optuna runs are treated as exploratory, not as formal final-model selection evidence. A post-hoc external check produced worse external F1 under the notebook comparison setup (Mel-CNN 0.385; Attention 0.503), but that comparison was not methodologically identical to the final manual-model evaluation because the Optuna external cell used a 0.5 threshold while the final models used validation-selected thresholds. The defensible conclusion is therefore narrower: finer in-domain tuning did not translate into a clear external-generalization improvement. The manually tuned checkpoints remain the final reported models.

3.6 Decision thresholds and evaluation metrics

For each final manual-grid model, the validation probabilities were swept over 200 thresholds from 0.01 to 0.99. The threshold minimizing the observed 1:5 business cost on validation was frozen and then applied to the SONICS test set and the external benchmark. Standard model comparison focused on precision, recall, F1, confusion-matrix counts, false-positive rate (FPR), false-negative rate (FNR), and the normalized business cost. The final report does not attach the notebook bootstrap confidence intervals to the threshold-optimized F1 values because those intervals were computed earlier at the default argmax/0.5 operating point.

4. Results

4.1 Threshold selection: theory versus observed calibration

The theoretically cost-optimal threshold of 0.167 was not universally effective. It improved Mel-CNN relative to the default 0.5 threshold, but it increased cost for Attention and catastrophically over-triggered the LSTM, which labeled almost all real test examples as fake. The validation-selected operating point had the lowest observed test cost for every model.

Model	Default cost (0.5)	Theory cost (0.167)	Val-selected threshold	Final test cost
Mel-CNN	0.238	0.197	0.2710	0.179
Attention / MERT	0.042	0.055	0.2809	0.034
Baseline LSTM	0.076	0.484	0.5763	0.069

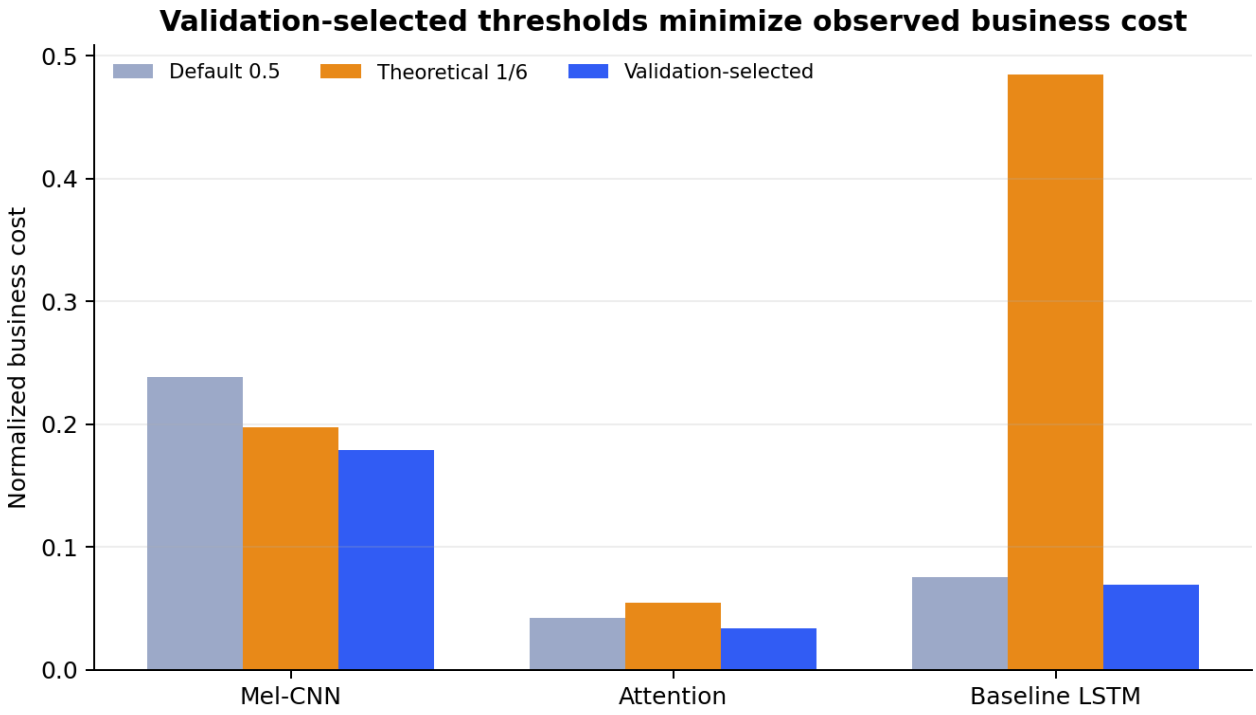


Figure 1. Normalized SONICS test business cost under default, theoretical, and validation-selected thresholds.

For the LSTM, the 0.167 threshold produced 718 false positives and zero false negatives, reducing precision to 0.505 and raising normalized cost to 0.484. By contrast, its validation-selected threshold of 0.5763 produced 67 false positives and seven false negatives, with cost 0.069. This is a direct example of why a cost-derived threshold is only valid when the score behaves like a calibrated probability.

4.2 Primary benchmark: SONICS held-out test set

The primary evaluation used all 1,482 held-out SONICS test songs, one fixed 10-second crop per song, and each model's validation-selected threshold. Attention/MERT was the strongest in-domain model by both F1 and weighted cost. The LSTM was a close second, while Mel-CNN was materially weaker.

Model	Precision	Recall	F1	Business cost	FP	FN
Baseline LSTM (scratch)	0.915	0.990	0.951	0.069	67	7
Mel-CNN (PANNs transfer)	0.830	0.967	0.893	0.179	145	24
Attention (MERT transfer)	0.948	0.997	0.972	0.034	40	2

Primary-result interpretation

On SONICS-like data, MERT provided the best combination of very high fake recall (99.7%) and controlled false positives. Under the project cost function, its two false negatives were especially valuable because a missed fake counted five times more than a false alarm.

4.3 Secondary benchmark: external generalization

The external benchmark told a substantially different story. All three models degraded, but the ranking flipped: the scratch LSTM became the strongest generalizer, while the two pretrained-backbone models fell to similar F1 values near 0.52-0.53. Because the external set contains unseen real-music sources and different AI-generation sources, this result measures robustness to domain shift rather than another sample from the SONICS distribution.

Model	Precision	Recall	F1	Business cost	FP	FN
Baseline LSTM (scratch)	0.708	0.658	0.682	0.956	560	703
Mel-CNN (PANNs transfer)	0.482	0.571	0.523	1.332	1,262	883
Attention (MERT transfer)	0.799	0.399	0.532	1.499	206	1,237

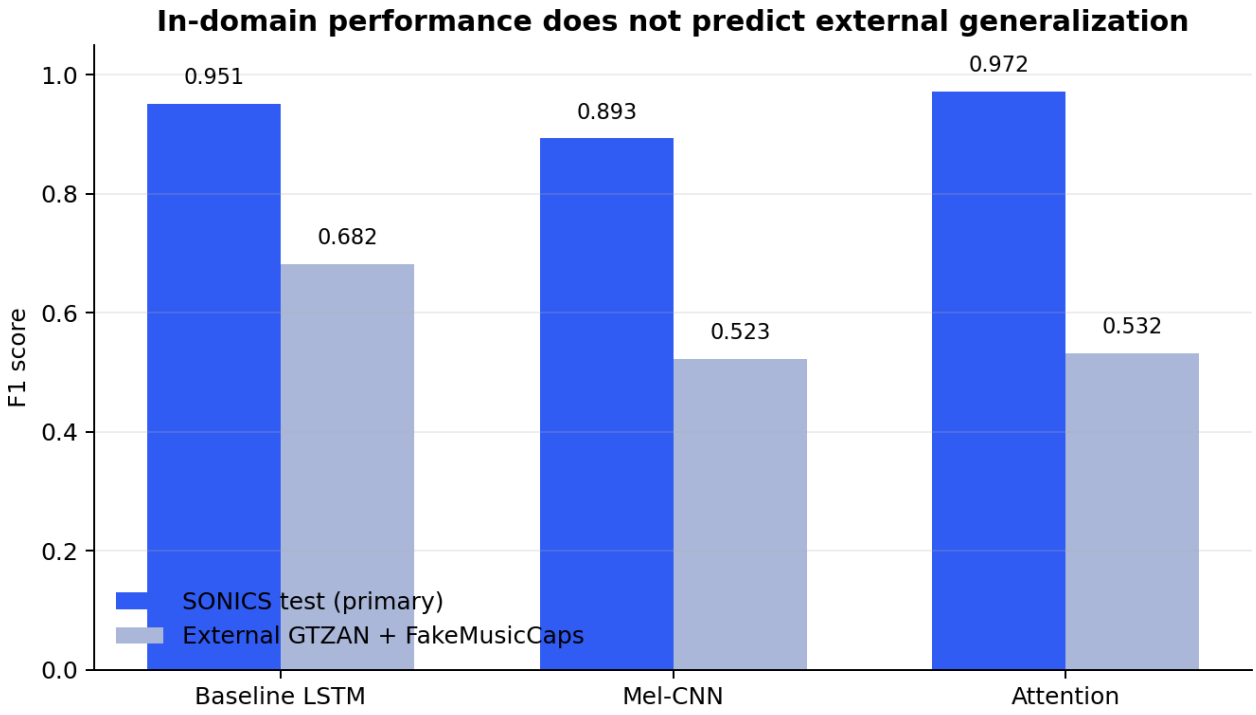


Figure 2. F1 on the primary SONICS test set versus the external GTZAN + FakeMusicCaps benchmark.

The normalized cost values on the external benchmark can exceed 1 because each false negative contributes five cost units before division by N. They should therefore be interpreted as weighted error cost, not as a probability or ordinary error rate.

4.4 Error-type breakdown on unseen data

Separating the real-only GTZAN side from the fake-only FakeMusicCaps side shows that the models did not simply become uniformly worse. They failed in different directions.

Model	GTZAN FPR: real -> fake	FakeMusicCaps FNR: fake -> real	External pattern
Baseline LSTM	25.4%	34.2%	Most balanced; best combined external F1 and cost
Mel-CNN	57.2%	42.9%	High error on both sides; weak overall generalization
Attention / MERT	9.3%	60.1%	Conservative: protects real music but misses unfamiliar fakes

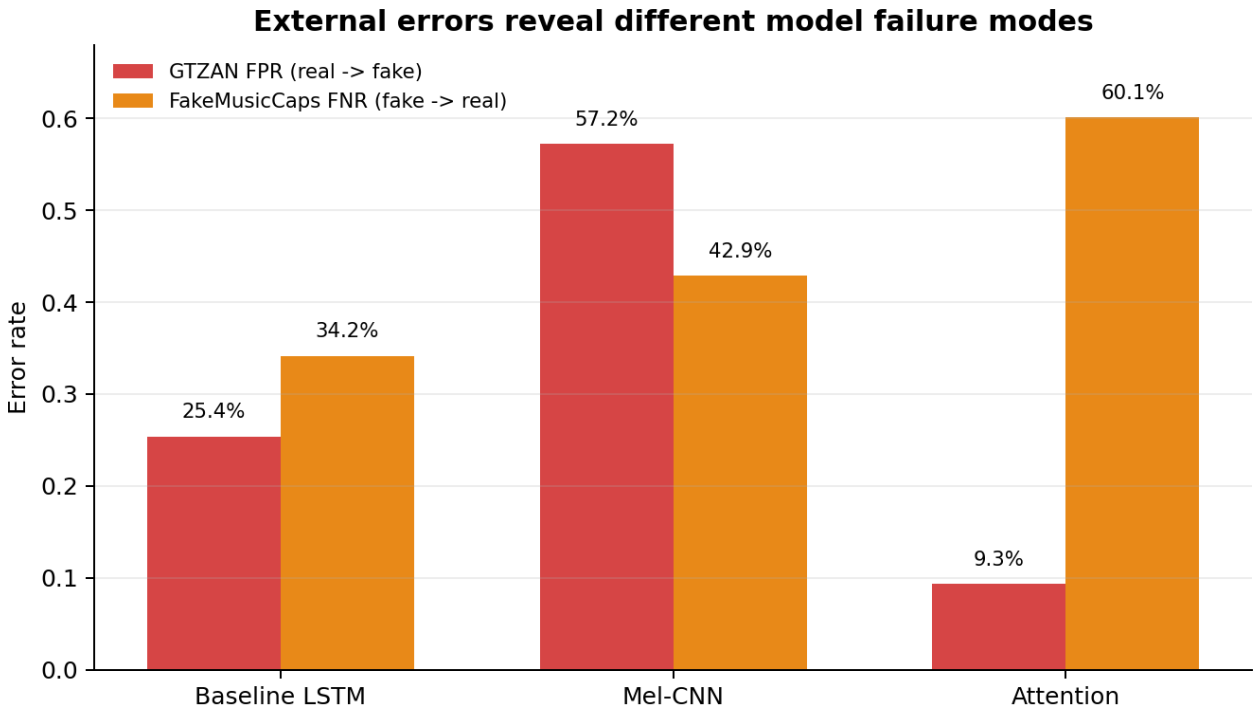


Figure 3. External false-positive and false-negative rates reveal distinct model operating characteristics.

Attention/MERT had the lowest GTZAN false-positive rate by a wide margin, incorrectly flagging only 9.3% of unfamiliar real-music segments. Its weakness was the opposite error: it missed 60.1% of FakeMusicCaps clips. Mel-CNN showed poor transfer on both classes, especially an unusually high 57.2% false-positive rate on GTZAN. The LSTM was not best on either single error axis, but its more balanced behavior produced the best combined external performance.

5. Discussion

5.1 Best in-domain model was not the best generalizer

The most important result is the ranking reversal between SONICS and the external benchmark. On SONICS, the MERT-based model appeared dominant: F1 0.972 and cost 0.034. On external data, its F1 fell to 0.532 and its cost rose to 1.499. The LSTM moved in the opposite relative direction: although it was weaker than MERT in-domain, it retained substantially more useful performance externally and achieved F1 0.682.

This suggests that the pretrained music representation captured highly discriminative cues for the SONICS distribution but did not reliably transfer those cues to unfamiliar AI generators. The scratch LSTM may have learned a less specialized set of spectral-temporal features. That interpretation is plausible rather than proven; the external benchmark also changes source, content distribution and recording characteristics, so a controlled generator-holdout experiment would be needed to isolate the cause.

5.2 Distinct failure modes matter more than one blended metric

The GTZAN/FakeMusicCaps breakdown shows why a single F1 score can hide operationally important behavior. Attention is comparatively safe for human creators because it rarely labels GTZAN music as AI, but it performs poorly if the business objective is to catch unfamiliar synthetic music. Mel-CNN is risky on both sides. The LSTM is the most balanced, which is why it wins the combined external benchmark despite not being best on either isolated error rate.

For a real deployment, the preferred model would therefore depend on the risk policy. A platform whose dominant concern is wrongful accusation of human creators might value Attention's low external FPR, whereas the stated 1:5 cost assumption strongly penalizes its 60.1% external FNR. Under the project's own business objective, the LSTM is the more defensible external candidate among the three tested models.

5.3 Theoretical cost thresholds require calibrated scores

The threshold experiment provides a clean theory-versus-practice result. The 1:5 cost ratio implies a 0.167 decision threshold only when the model score is a calibrated estimate of P(fake). The three networks clearly did not share that calibration. For the LSTM, 0.167 was far too low; for Attention, it increased false positives enough to worsen cost; only Mel-CNN benefited relative to 0.5. A simple validation sweep found a better operating point for every model.

This result has practical importance beyond the specific architectures: deployment thresholds should be treated as model-specific operating parameters, selected on representative validation data under the actual business loss. Future work should add explicit probability-calibration analysis rather than assuming that softmax scores are directly comparable across models.

5.4 What the Optuna experiment does and does not show

Optuna found configurations with lower SONICS validation business cost than the manual grid, but those gains did not translate into better external results in the exploratory check. This is consistent with increased specialization to SONICS, but the post-hoc comparison is not strong enough to establish causality because the threshold treatment differed between the Optuna and final-model external cells. The reliable takeaway is that more aggressive in-domain optimization did not solve the generalization problem. External robustness therefore needs to be designed and measured directly rather than inferred from progressively better validation scores.

6. Limitations and Methodological Caveats

Several limitations should be considered when interpreting the reported numbers. They do not invalidate the core comparisons, but they define what can and cannot be concluded from the current experiment.

Unequal augmentation across architectures. Dynamic noise/bitrate augmentation was applied to the LSTM waveform path, but Mel-CNN and MERT used cached embeddings from unaugmented audio. The transfer models therefore did not receive the same training augmentation. A rerun with augmentation applied before embedding extraction is the highest-priority implementation correction.

External source and label are partially confounded. GTZAN supplies the real class and FakeMusicCaps supplies the fake class. They were not generated from matched prompts, genres or source conditions. A model can therefore respond to dataset-specific cues as well as AI-generation cues. The external benchmark is a useful stress test, but not a perfectly controlled causal test of generator invariance.

GTZAN evaluation items are not all independent recordings. The 2,205 GTZAN items come from 735 usable source recordings split into three segments each. Segment-level metrics therefore contain within-song dependence. Song-level aggregation or clustered confidence intervals would be more statistically conservative.

Compute-constrained training subset. Hyperparameter search and final training used a balanced 2,100-song subset of the 6,915 SONICS training songs rather than the complete training split. This made experimentation feasible but leaves open how each architecture would scale with all available training songs.

One fixed crop per validation/test song. Final SONICS evaluation uses one deterministic 10-second crop per song. This creates a fair one-song/one-vote comparison, but performance could vary across different excerpts of the same recording. Multi-crop song-level aggregation would better characterize full-song robustness.

Supplementary Optuna comparison is exploratory. The Optuna test cell performed a test-side threshold sweep and its external cell used threshold 0.5, unlike the validation-selected thresholds used for the final models. Those numbers are therefore excluded from primary model selection and interpreted only qualitatively.

Final-threshold confidence intervals were not recomputed. The notebook bootstrap intervals were generated for default-threshold predictions before the final validation-selected thresholds were applied. To avoid mixing operating points, this report does not attach those intervals to the final F1 scores.

7. Engineering, Debugging, and Reproducibility

A significant part of the project was making the experimental pipeline reliable enough to support repeated training and evaluation in Google Colab with Drive-backed storage. The debugging process was evidence-driven: failures were traced from exceptions, file counts and timing behavior rather than handled by silent workarounds.

Issue encountered	Resolution used in the final pipeline
Drive I/O bottleneck	Selected clips were archived once on Drive and unpacked to Colab local disk for repeated training/embedding reads.
PANNs sample-rate mismatch	The default 32 kHz setup was replaced with the matching Cnn14_16k checkpoint/configuration for the 16 kHz pipeline.
Stale/corrupted archives and missing local clips	Archives were validated before reuse; corrupted/zero-byte archives were rebuilt, and missing selected clips could be replaced with another clip from the same song.
Checkpoint/loading compatibility	The training code saved full model/optimizer states and used explicit loading behavior for checkpoint formats that required it.
Long transfer-backbone computation	Frozen PANNs/MERT embeddings were cached and reused across head-training HPO runs.

Each run saved its configuration and checkpoints, and the reporting blocks saved metric tables and per-item probabilities/predictions. The final checkpoints used in the report are baseline_lstm_hpo_2, melcnn_hpo_lr0.01, and attention_hpo_lr0.01.

8. Conclusions and Recommended Next Steps

The project successfully built a reproducible deep-learning pipeline for AI-music detection, corrected a key unit-of-analysis problem during execution, compared three materially different architectures, incorporated asymmetric business costs into the decision rule, and tested generalization on unseen sources. The final results support four main conclusions:

- In-domain performance alone is not a reliable proxy for deployment robustness. MERT was the SONICS leader, yet the LSTM was the strongest external generalizer.
- Failure modes are architecture-specific. Attention/MERT was highly conservative on unfamiliar real music but missed most unfamiliar fake music; Mel-CNN failed strongly on both sides; the LSTM was more balanced.
- Empirical, validation-driven thresholds were better than both the default 0.5 threshold and the theoretical 1/6 cost threshold for all three models under the stated loss function.
- More in-domain tuning did not solve domain shift. The supplementary Optuna search improved validation cost but provided no convincing external benefit.

If the task were extended toward a deployment-quality detector, the next experiments should first fix augmentation parity across architectures, then evaluate on additional generators and matched real/fake content, aggregate multiple crops at the song level, and explicitly calibrate model probabilities before threshold selection. Training on the full SONICS training split and testing generator-held-out or source-held-out folds would help determine whether the LSTM's external advantage is stable or specific to the current benchmark.

Final recommendation

For SONICS-like data, Attention/MERT is the strongest model. For unfamiliar-source robustness under the current external benchmark and 1:5 miss-cost assumption, the Baseline LSTM is the strongest of the tested models - but its external F1 of 0.682 is not high enough to treat the detector as deployment-ready without further validation.

Appendix A. Final configurations and exact operating points

Model	Checkpoint	Backbone / hidden	Dropout	LR	Epochs	Final threshold
Baseline LSTM	baseline_lstm_hpo_2	2-layer LSTM, hidden 128	0.5	5e-4	5	0.5763
Mel-CNN	melcnn_hpo_lr0.01	Frozen PANNs Cnn14_16k	0.3	0.01	10	0.2710
Attention / MERT	attention_hpo_lr0.01	Frozen MERT-v1-95M	0.3	0.01	10	0.2809

Appendix B. Major execution-time changes from the original plan

Original or early design	Final executed approach	Reason / consequence
Flat clip-pool sampling	Song-level sampling; one song = one example, random candidate clip during training	Avoids over-weighting correlated clips and fixes the effective unit of analysis.
General PANNs transfer setup	Cnn14_16k checkpoint/configuration	Matches the project's 16 kHz audio instead of silently using a 32 kHz assumption.
MusicCaps + FakeMusicCaps binary benchmark, GTZAN separate	GTZAN real + FakeMusicCaps fake combined benchmark, with source-specific breakdown	Provides full external precision/recall/F1 while still exposing real-only FPR and fake-only FNR.
Manual HPO only	Manual grid as primary + supplementary Optuna search	Tests whether finer in-domain tuning improves generalization; final report retains manual checkpoints.
Uniform training augmentation planned	Augmentation reached LSTM but not cached transfer embeddings	Known implementation gap; limits strict architecture-to-architecture comparability.

Project materials and implementation references

- Group 10. Project Architecture: Blocks, Interfaces, and Colab Strategy - Version 3 (August 2026).
- Group 10. Real vs. AI-Generated Audio: Main Pipeline (v2), Jupyter/Colab notebook.
- Group 10. preliminary_data_preparation, Jupyter/Colab notebook.
- Group 10. Experimental Results and Key Findings & Conclusions, project presentation.
- SONICS dataset resources referenced in the project notebook: github.com/awsaf49/sonics and Kaggle SONICS dataset page.
- PANNs Cnn14_16k checkpoint referenced by the implementation; MERT model identifier: m-a-p/MERT-v1-95M.